

PRACTICE EXPERIMENTS UNIT 2

CSA6501-GENAI

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A program that stores and displays the Zero-shot, One-shot, and Few-shot prompts:

```
zero_shot = ""
```

Generate a professional product description for a Smart Fitness Watch.

Highlight its features such as heart rate monitoring, sleep tracking,

GPS, waterproof design, long battery life, and smartphone connectivity.

```
""
```

```
one_shot = ""
```

Example:

Product: Wireless Bluetooth Earbuds

Description:

Experience premium sound quality with Bluetooth Earbuds featuring noise cancellation, ergonomic design, and long battery life.

Now generate a similar product description for a Smart Fitness Watch.

```
""
```

```
few_shot = ""
```

Example 1:

Product: Wireless Bluetooth Earbuds

Description: High-quality earbuds with noise cancellation.

Example 2:

Product: Smart Home Camera

Description: HD security camera with motion detection.

Example 3:

Product: Portable Power Bank

Description: Fast-charging power bank with high capacity.

Now generate a product description for a Smart Fitness Watch.

```
""
```

```

print("===== ZERO-SHOT PROMPT =====\n")

print(zero_shot)

print("\n===== ONE-SHOT PROMPT =====\n")

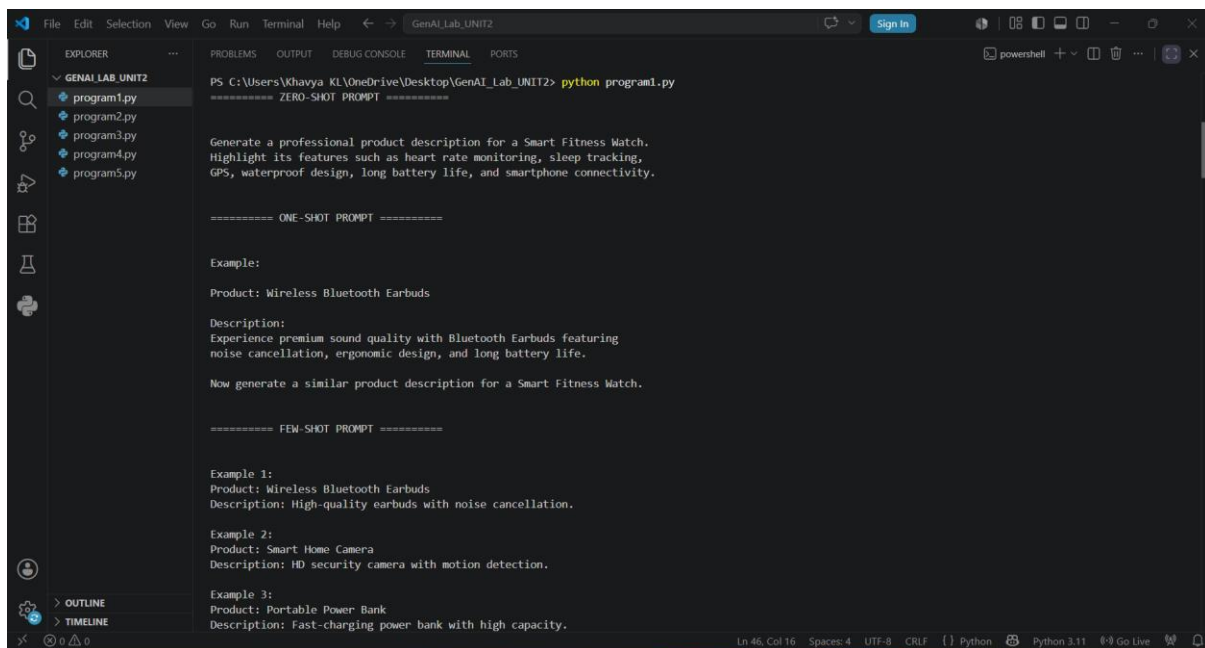
print(one_shot)

print("\n===== FEW-SHOT PROMPT =====\n")

print(few_shot)

```

OUTPUT:



```

PS C:\Users\Khavya KL\OneDrive\Desktop\GenAI_Lab_UNIT2> python program1.py
===== ZERO-SHOT PROMPT =====

Generate a professional product description for a Smart Fitness Watch.
Highlight its features such as heart rate monitoring, sleep tracking,
GPS, waterproof design, long battery life, and smartphone connectivity.

===== ONE-SHOT PROMPT =====

Example:

Product: Wireless Bluetooth Earbuds

Description:
Experience premium sound quality with Bluetooth Earbuds featuring
noise cancellation, ergonomic design, and long battery life.

Now generate a similar product description for a Smart Fitness Watch.

===== FEW-SHOT PROMPT =====

Example 1:
Product: Wireless Bluetooth Earbuds
Description: High-quality earbuds with noise cancellation.

Example 2:
Product: Smart Home Camera
Description: HD security camera with motion detection.

Example 3:
Product: Portable Power Bank
Description: Fast-charging power bank with high capacity.

```

Design Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog on "Applications of Artificial Intelligence in Healthcare.":

zero_shot = ""

Write a 200-word blog on 'Applications of Artificial Intelligence in Healthcare'.

Explain how AI is used in disease diagnosis, medical imaging, drug discovery, robotic surgery, patient monitoring, and personalized medicine.

Use a professional and informative tone.

""

one_shot = ""

Example:

Topic: Applications of Artificial Intelligence in Education

Blog:

Artificial Intelligence is transforming education by enabling personalized learning, automated grading, intelligent tutoring systems, and virtual classrooms.

It improves learning experiences and enhances educational outcomes.

Now write a similar 200-word blog on

'Applications of Artificial Intelligence in Healthcare'.

""

few_shot = ""

Example 1:

Topic: Artificial Intelligence in Education

Blog:

AI improves personalized learning, automated assessments, and student engagement.

Example 2:

Topic: Artificial Intelligence in Agriculture

Blog:

AI helps monitor crops, detect plant diseases, and improve farming efficiency.

Example 3:

Topic: Artificial Intelligence in Banking

Blog:

AI enhances fraud detection, customer service, and financial forecasting.

Now write a 200-word blog on

'Applications of Artificial Intelligence in Healthcare'.

```
"""
```

```
print("===== ZERO-SHOT PROMPT =====\n")
```

```
print(zero_shot)
```

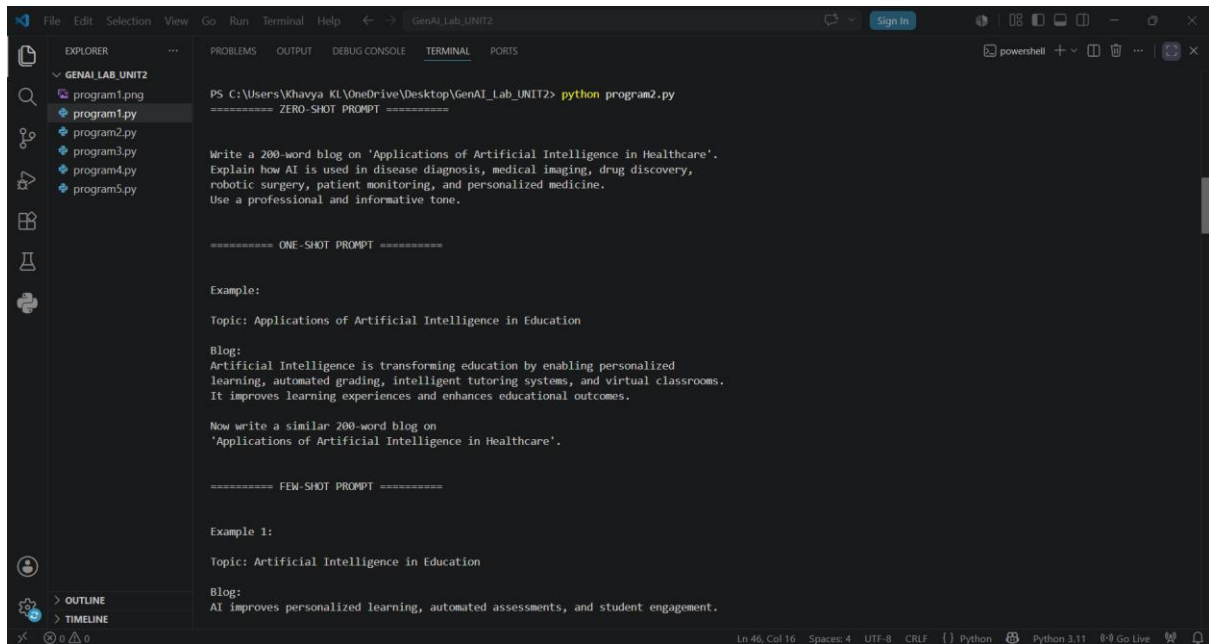
```
print("\n===== ONE-SHOT PROMPT =====\n")
```

```
print(one_shot)
```

```
print("\n===== FEW-SHOT PROMPT =====\n")
```

```
print(few_shot)
```

OUTPUT:



```
PS C:\Users\Khavya KL\OneDrive\Desktop\GenAI_Lab_UNIT2> python program2.py
===== ZERO-SHOT PROMPT =====

Write a 200-word blog on 'Applications of Artificial Intelligence in Healthcare'.
Explain how AI is used in disease diagnosis, medical imaging, drug discovery,
robotic surgery, patient monitoring, and personalized medicine.
Use a professional and informative tone.

===== ONE-SHOT PROMPT =====

Example:

Topic: Applications of Artificial Intelligence in Education

Blog:
Artificial Intelligence is transforming education by enabling personalized
learning, automated grading, intelligent tutoring systems, and virtual classrooms.
It improves learning experiences and enhances educational outcomes.

Now write a similar 200-word blog on
'Applications of Artificial Intelligence in Healthcare'.

===== FEW-SHOT PROMPT =====

Example 1:

Topic: Artificial Intelligence in Education

Blog:
AI improves personalized learning, automated assessments, and student engagement.
```

Design Zero-shot, One-shot, and Few-shot prompts to summarize the following article into 50 words. Compare the generated summaries based on accuracy, completeness, and readability.:

article = ""

Artificial Intelligence (AI) is transforming industries by automating repetitive tasks, improving decision-making, and increasing efficiency. AI applications include healthcare, education, finance, agriculture, transportation, and manufacturing. Machine learning, deep learning, and natural language processing enable AI systems to analyze large volumes of data, recognize patterns, and make accurate predictions. Although AI offers numerous benefits, challenges such as data privacy, ethical concerns, and job displacement must also be addressed for responsible and sustainable adoption.

""

zero_shot = f""

Summarize the following article in exactly 50 words.

Article:

{article}

""

one_shot = f""

Example:

Article:

Online learning allows students to study anytime and anywhere using digital platforms.

Summary:

Online learning provides flexible and accessible education through digital technology, helping students learn conveniently from any location.

Now summarize the following article in exactly 50 words.

Article:

{article}

""

few_shot = f""

Example 1:

Article:

Solar energy is a clean and renewable source of power.

Summary:

Solar energy reduces pollution by providing sustainable electricity from sunlight.

Example 2:

Article:

Electric vehicles reduce fuel consumption and carbon emissions.

Summary:

Electric vehicles support eco-friendly transportation by lowering pollution and fuel dependency.

Example 3:

Article:

Cloud computing enables remote storage and computing services.

Summary:

Cloud computing offers scalable, secure, and accessible computing resources over the internet.

Now summarize the following article in exactly 50 words.

Article:

{article}

"""

```
print("===== ZERO-SHOT PROMPT =====\n")
```

```
print(zero_shot)
```

```
print("\n===== ONE-SHOT PROMPT =====\n")
```

```
print(one_shot)
```

```
print("\n===== FEW-SHOT PROMPT =====\n")
```

```
print(few_shot)
```

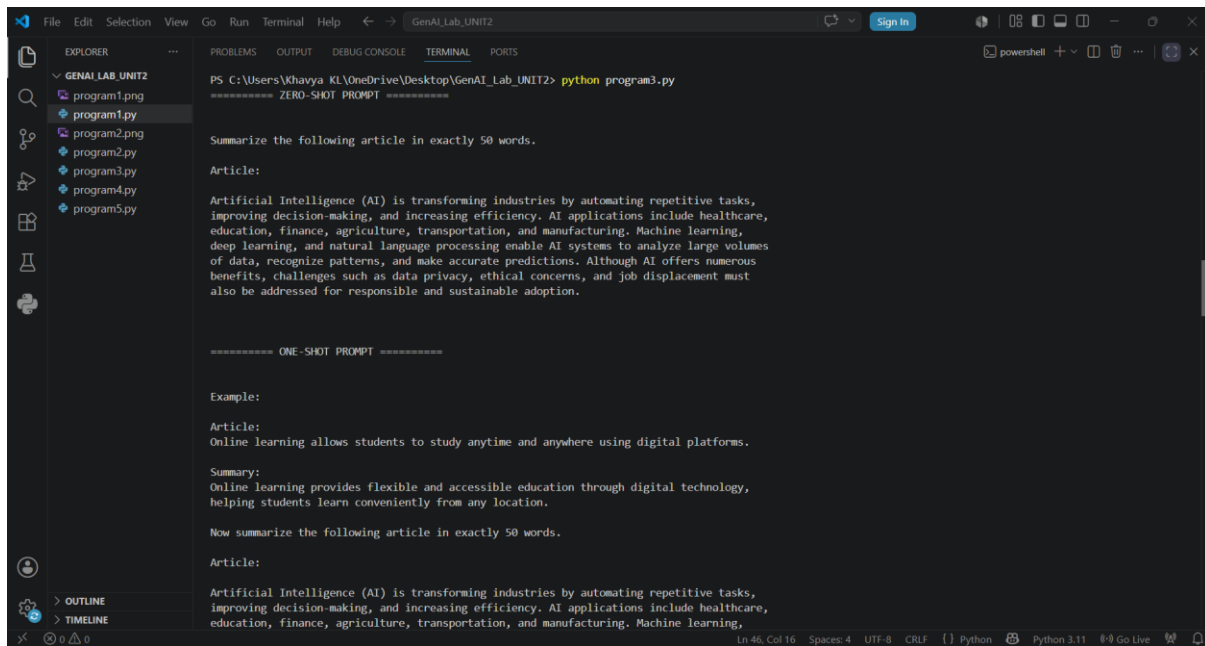
```
print("\n===== COMPARISON =====\n")
```

```
print("Zero-shot : Good accuracy, moderate completeness, readable summary.")
```

```
print("One-shot : Better accuracy, improved completeness, clearer readability.")
```

```
print("Few-shot : Highest accuracy, excellent completeness, and best readability.")
```

OUTPUT:



```
PS C:\Users\Khavya KL\OneDrive\Desktop\GenAI_Lab_UNIT2> python program3.py
===== ZERO-SHOT PROMPT =====

Summarize the following article in exactly 50 words.

Article:

Artificial Intelligence (AI) is transforming industries by automating repetitive tasks,
improving decision-making, and increasing efficiency. AI applications include healthcare,
education, finance, agriculture, transportation, and manufacturing. Machine learning,
deep learning, and natural language processing enable AI systems to analyze large volumes
of data, recognize patterns, and make accurate predictions. Although AI offers numerous
benefits, challenges such as data privacy, ethical concerns, and job displacement must
also be addressed for responsible and sustainable adoption.

===== ONE-SHOT PROMPT =====

Example:

Article:
Online learning allows students to study anytime and anywhere using digital platforms.

Summary:
Online learning provides flexible and accessible education through digital technology,
helping students learn conveniently from any location.

Now summarize the following article in exactly 50 words.

Article:

Artificial Intelligence (AI) is transforming industries by automating repetitive tasks,
improving decision-making, and increasing efficiency. AI applications include healthcare,
education, finance, agriculture, transportation, and manufacturing. Machine learning,
```

Design Zero-shot, One-shot, and Few-shot prompts to generate a professional email requesting leave due to illness. Compare the generated emails based on tone, grammar, formatting, and completeness.:

```
zero_shot = ""
```

Write a professional email requesting two days of leave due to illness.

The email should include a suitable subject, a polite greeting, the reason for leave, the leave duration, and a formal closing.

```
""
```

```
one_shot = ""
```

Example:

Subject: Leave Request for Personal Work

Dear Manager,

I would like to request one day's leave due to personal commitments.

I have completed my pending tasks and will resume work tomorrow.

Thank you for your understanding.

Regards,

Rahul

Now generate a similar professional email requesting two days of leave due to illness.

"""

few_shot = """

Example 1:

Subject: Leave Request for Family Function

Dear Manager,

I request one day's leave to attend a family function.

Thank you for your consideration.

Regards,

Amit

Example 2:

Subject: Leave Request for Medical Appointment

Dear Manager,

I kindly request leave for a medical appointment tomorrow.

Thank you for your support.

Regards,

Sneha

Example 3:

Subject: Leave Request for Personal Work

Dear Manager,

I would like to request leave due to personal reasons.

Thank you for your understanding.

Regards,

Karan

Now generate a professional email requesting two days of leave due to illness.

"""

```
print("===== ZERO-SHOT PROMPT =====\n")
```

```
print(zero_shot)
```

```
print("\n===== ONE-SHOT PROMPT =====\n")
```

```
print(one_shot)
```

```
print("\n===== FEW-SHOT PROMPT =====\n")
```



```
print(few_shot)
```

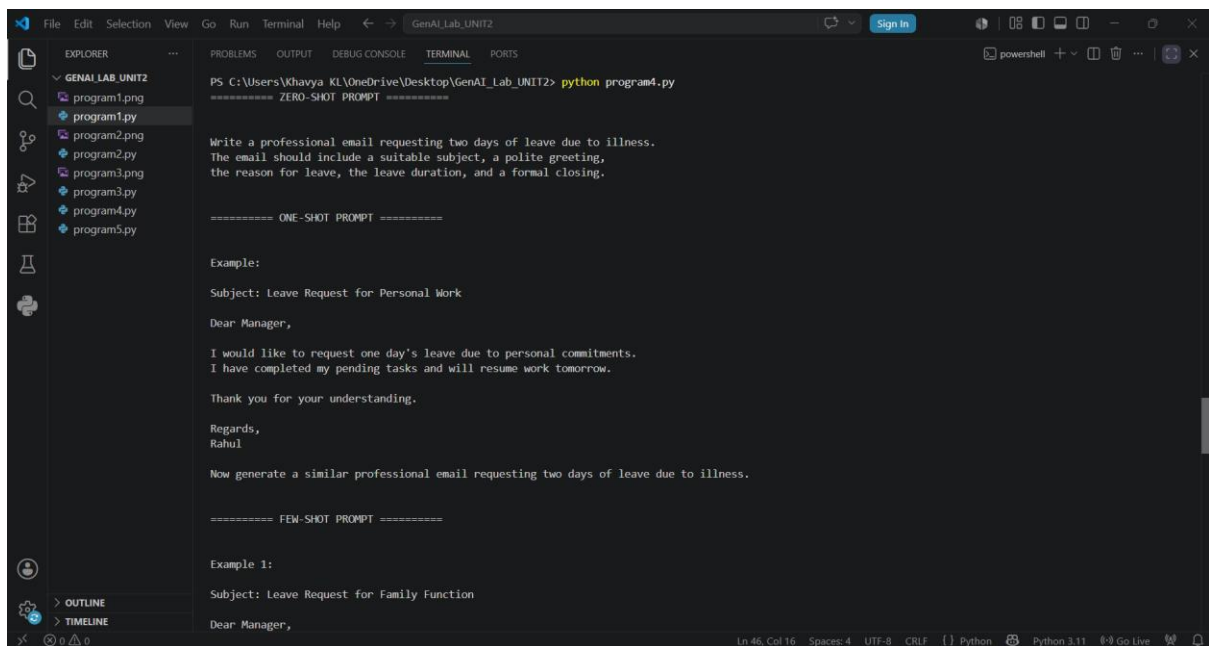
```
print("\n===== COMPARISON =====\n")
```

```
print("Zero-shot : Professional tone, grammatically correct, basic formatting, includes essential details.")
```

```
print("One-shot : More polished tone, improved grammar, better email structure, more complete information.")
```

```
print("Few-shot : Highly professional tone, excellent grammar, proper formatting, and comprehensive content.")
```

OUTPUT:



```
PS C:\Users\Khavya KI\OneDrive\Desktop\GenAI_Lab_UNIT2> python program4.py
===== ZERO-SHOT PROMPT =====

Write a professional email requesting two days of leave due to illness.
The email should include a suitable subject, a polite greeting,
the reason for leave, the leave duration, and a formal closing.

===== ONE-SHOT PROMPT =====

Example:
Subject: Leave Request for Personal Work

Dear Manager,

I would like to request one day's leave due to personal commitments.
I have completed my pending tasks and will resume work tomorrow.

Thank you for your understanding.

Regards,
Rahul

Now generate a similar professional email requesting two days of leave due to illness.

===== FEW-SHOT PROMPT =====

Example 1:
Subject: Leave Request for Family Function

Dear Manager,
```

Design Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop.

zero_shot = ""

Generate a promotional social media post for an AI Workshop.

The post should be engaging, attractive, and encourage students to register.

Include workshop highlights such as hands-on sessions, expert speakers, certificate of participation, date, venue, and registration details.

""

one_shot = ""

Example:

Event: Python Programming Workshop

Social Media Post:

 Join our Python Programming Workshop!

Learn Python from industry experts through interactive hands-on sessions.

Receive a participation certificate and improve your programming skills.

 Date: 20th August

 Venue: Seminar Hall

 Register Now!

Now generate a similar promotional social media post for an AI Workshop.

""

few_shot = ""

Example 1:

Event: Data Science Workshop

Post:

 Explore the world of Data Science through practical sessions, expert guidance, and real-world case studies.

Example 2:

Event: Cyber Security Workshop

Post:

 Learn ethical hacking, cyber threats, and security techniques

from experienced professionals.

Example 3:

Event: Cloud Computing Workshop

Post:

☁ Discover cloud technologies with hands-on demonstrations, industry experts, and certification.

Now generate a promotional social media post for an AI Workshop.

```
"""
```

```
print("===== ZERO-SHOT PROMPT =====\n")
```

```
print(zero_shot)
```

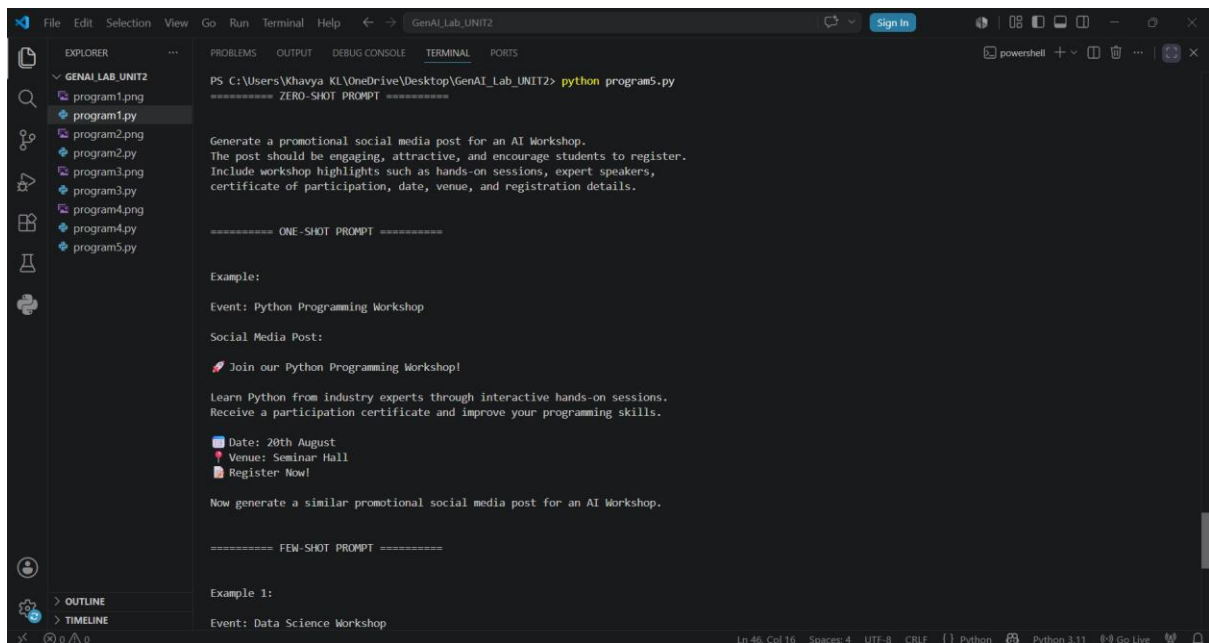
```
print("\n===== ONE-SHOT PROMPT =====\n")
```

```
print(one_shot)
```

```
print("\n===== FEW-SHOT PROMPT =====\n")
```

```
print(few_shot)
```

OUTPUT:



```
PS C:\Users\Khavya KI\OneDrive\Desktop\GenAI_Lab_UNIT2> python program5.py
===== ZERO-SHOT PROMPT =====

Generate a promotional social media post for an AI Workshop.
The post should be engaging, attractive, and encourage students to register.
Include workshop highlights such as hands-on sessions, expert speakers,
certificate of participation, date, venue, and registration details.

===== ONE-SHOT PROMPT =====

Example:

Event: Python Programming Workshop

Social Media Post:

🔥 Join our Python Programming Workshop!

Learn Python from industry experts through interactive hands-on sessions.
Receive a participation certificate and improve your programming skills.

📅 Date: 20th August
📍 Venue: Seminar Hall
📌 Register Now!

Now generate a similar promotional social media post for an AI Workshop.

===== FEW-SHOT PROMPT =====

Example 1:

Event: Data Science Workshop
```